

DBMS – PRODUCT AND CATEGORY MANAGEMENT

Project Title:

SAVANA ONLINE E-COMMERCE DATABASE MANAGEMENT SYSTEM

1.Design Database

The Savana Online E-Commerce Database Management System is designed to manage products and their categories in an online fashion and lifestyle shopping platform. The system stores product information such as product name, seller, category, price, stock, brand, color, and size. The Category entity stores information about different categories available in the Savana online shopping platform.

1.1 Category Table

Attribute	Data Type	Constraint
Category_ID	NUMBER(5)	Primary Key
Category_Name	VARCHAR2(50)	UNIQUE
Description	VARCHAR(100)	-

1.2 Product Table

Attribute	Data Type	Constraint
Product_ID	NUMBER(5)	Primary Key
Product_Name	VARCHAR2(100)	NOT NULL
Seller_ID	NUMBER(5)	Foreign Key
Category_ID	NUMBER(5)	Foreign Key
Price	NUMBER(10,2)	NOT NULL
Stock	NUMBER(5)	CHECK
Brand	VARCHAR2(50)	-
Colour	VARCHAR2(30)	-
Size	VARCHAR2(20)	-

2. Primary and Foreign Key Relationships

The relationship between the Category and Product tables is established using CATEGORY_ID.

- CATEGORY.CATEGORY_ID → Primary Key
- PRODUCT.PRODUCT_ID → Primary Key
- PRODUCT.CATEGORY_ID → Foreign Key referencing CATEGORY.CATEGORY_ID
- Relationship is One-to-Many (1:M).
- One category can contain many products, while each product belongs to a particular category.

3. Store Product Details and Table Creation

- The Savana database stores product details required for online shopping. These details include product name, category, price, stock quantity, brand, color, and size.
- The Category and Product tables are created using SQL with suitable primary key and foreign key constraints.

3.1 Create Category Table

```
SQL> CREATE TABLE SAVANA_CATEGORY (  
2     CATEGORY_ID NUMBER(5) PRIMARY KEY,  
3     CATEGORY_NAME VARCHAR2(50) NOT NULL,  
4     DESCRIPTION VARCHAR2(100)  
5 );
```

Table created.

3.2 Create product Table

```
SQL> CREATE TABLE SAVANA_PRODUCT (  
2     PRODUCT_ID NUMBER(5) PRIMARY KEY,  
3     PRODUCT_NAME VARCHAR2(100) NOT NULL,  
4     CATEGORY_ID NUMBER(5),  
5     PRICE NUMBER(10,2) NOT NULL,  
6     STOCK NUMBER(5),  
7     BRAND VARCHAR2(50),  
8     COLOR VARCHAR2(30),  
9     PRODUCT_SIZE VARCHAR2(20),  
10    CONSTRAINT FK_SAVANA_PRODUCT_CATEGORY  
11    FOREIGN KEY (CATEGORY_ID)  
12    REFERENCES SHOP_CATEGORY(CATEGORY_ID)  
13 );
```

Table created.

\

4.Product Insertion, Updating and Deletion

4.1 Category Records

```
SQL> INSERT INTO SHOP_CATEGORY
  2 (CATEGORY_ID, CATEGORY_NAME, DESCRIPTION)
  3 VALUES (1, 'Electronics', 'Electronic products');

1 row created.

SQL>
SQL> INSERT INTO SHOP_CATEGORY
  2 (CATEGORY_ID, CATEGORY_NAME, DESCRIPTION)
  3 VALUES (2, 'Fashion', 'Fashion products');

1 row created.

SQL>
SQL> INSERT INTO SHOP_CATEGORY
  2 (CATEGORY_ID, CATEGORY_NAME, DESCRIPTION)
  3 VALUES (3, 'Home Appliances', 'Home appliance products');

1 row created.

SQL>
SQL> COMMIT;|
```

4.2 Product Records

```
SQL> INSERT INTO SHOP_PRODUCT
  2 (PRODUCT_ID, PRODUCT_NAME, CATEGORY_ID, PRICE, STOCK)
  3 VALUES (102, 'Smart Watch', 1, 2999, 15);

1 row created.

SQL>
SQL> INSERT INTO SHOP_PRODUCT
  2 (PRODUCT_ID, PRODUCT_NAME, CATEGORY_ID, PRICE, STOCK)
  3 VALUES (103, 'Cotton Shirt', 2, 899, 40);

1 row created.

SQL>
SQL> INSERT INTO SHOP_PRODUCT
  2 (PRODUCT_ID, PRODUCT_NAME, CATEGORY_ID, PRICE, STOCK)
  3 VALUES (104, 'Denim Jeans', 2, 1499, 30);

1 row created.

SQL>
SQL> INSERT INTO SHOP_PRODUCT
  2 (PRODUCT_ID, PRODUCT_NAME, CATEGORY_ID, PRICE, STOCK)
  3 VALUES (105, 'Electric Kettle', 3, 1299, 20);

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

4.3 Product Update

The price and stock quantity of the Wireless Headphones product are updated.

```
SQL>
SQL> UPDATE SHOP_PRODUCT
  2  SET PRICE = 1799,
  3      STOCK = 30
  4  WHERE PRODUCT_ID = 101;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.
```

4.4 Product Deletion

The Electric Kettle product is deleted from the Product table.

```
SQL> DELETE FROM SHOP_PRODUCT
  2  WHERE PRODUCT_ID = 105;

1 row deleted.

SQL>
SQL> COMMIT;

Commit complete.
```

5. Category-wise Product Reports

Category-wise reports are generated in the Savana database to display products according to their respective categories. These reports help the administrator to view and manage products available under each category.

5.1 Product Listing by Category

```
SQL> SELECT
2     C.CATEGORY_NAME,
3     P.PRODUCT_NAME,
4     P.PRICE,
5     P.STOCK
6 FROM SHOP_CATEGORY C
7 JOIN SHOP_PRODUCT P
8 ON C.CATEGORY_ID = P.CATEGORY_ID
9 ORDER BY C.CATEGORY_NAME;
```

CATEGORY_NAME

PRODUCT_NAME

PRICE	STOCK
Electronics	
Wireless Headphones	
1799	30

Electronics	
Smart Watch	
2999	15

CATEGORY_NAME

PRODUCT_NAME

PRICE	STOCK
Fashion	
Cotton Shirt	
899	40

Fashion	
Denim Jeans	

CATEGORY_NAME

PRODUCT_NAME

PRICE	STOCK
1499	30

5.2 Products in Fashion Category

```
SQL> SELECT
  2     P.PRODUCT_ID,
  3     P.PRODUCT_NAME,
  4     P.PRICE,
  5     P.STOCK
  6 FROM SHOP_PRODUCT P
  7 JOIN SHOP_CATEGORY C
  8 ON P.CATEGORY_ID = C.CATEGORY_ID
  9 WHERE C.CATEGORY_NAME = 'Fashion';
```

PRODUCT_ID		
PRODUCT_NAME		
	PRICE	STOCK
103		
Cotton Shirt	899	40
104		
Denim Jeans	1499	30

PRODUCT_ID		
PRODUCT_NAME		
	PRICE	STOCK

This query displays all products that belong to the Fashion category.

5.3 Product Count by Category

```
SQL>
SQL> SELECT
  2     C.CATEGORY_NAME,
  3     COUNT(P.PRODUCT_ID) AS PRODUCT_COUNT
  4 FROM SHOP_CATEGORY C
  5 LEFT JOIN SHOP_PRODUCT P
  6 ON C.CATEGORY_ID = P.CATEGORY_ID
  7 GROUP BY C.CATEGORY_NAME
  8 ORDER BY C.CATEGORY_NAME;
```

CATEGORY_NAME	PRODUCT_COUNT
Electronics	2
Fashion	2
Home Appliances	0

Conclusion

The Savana Online E-Commerce Database Management System demonstrates the design and implementation of the Product and Category entities. The system establishes primary and foreign key relationships between Category and Product, stores important product details such as product name, category, price, stock, brand, color, and size, and performs insertion, updating, and deletion operations.

The system also generates category-wise product reports, which helps in organizing and managing products efficiently. Thus, the Product and Category database design provides a structured and reliable foundation for the Savana online e-commerce platform.